

Core Architectures of Generative AI



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Core Architectures

Transformers



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**Generative Adversarial Networks
(GANs)**



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Variational Autoencoders (VAEs)



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Diffusion Models



Generative AI Models Compared

MODEL	YEAR INTRODUCED	STRENGTHS	COMMON USE CASES
Transformers	2017	Text and multimodal generation	ChatGPT, BERT, Gemini
Generative Adversarial Networks (GANs)	2014	High-quality image synthesis	StyleGAN, Deepfakes
Variational Autoencoders (VAEs)	2013	Controlled variation	MNIST, bio/medical data
Diffusion Models	2020	Ultra-high quality images and videos	DALL-E 3, Sora, Stable Diffusion, Midjourney





Generative Adversarial Networks (GANs) Explained



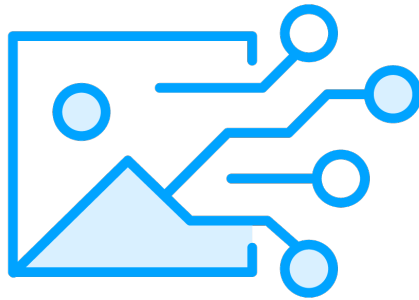




Real or fake?



The Two Parts of a GAN

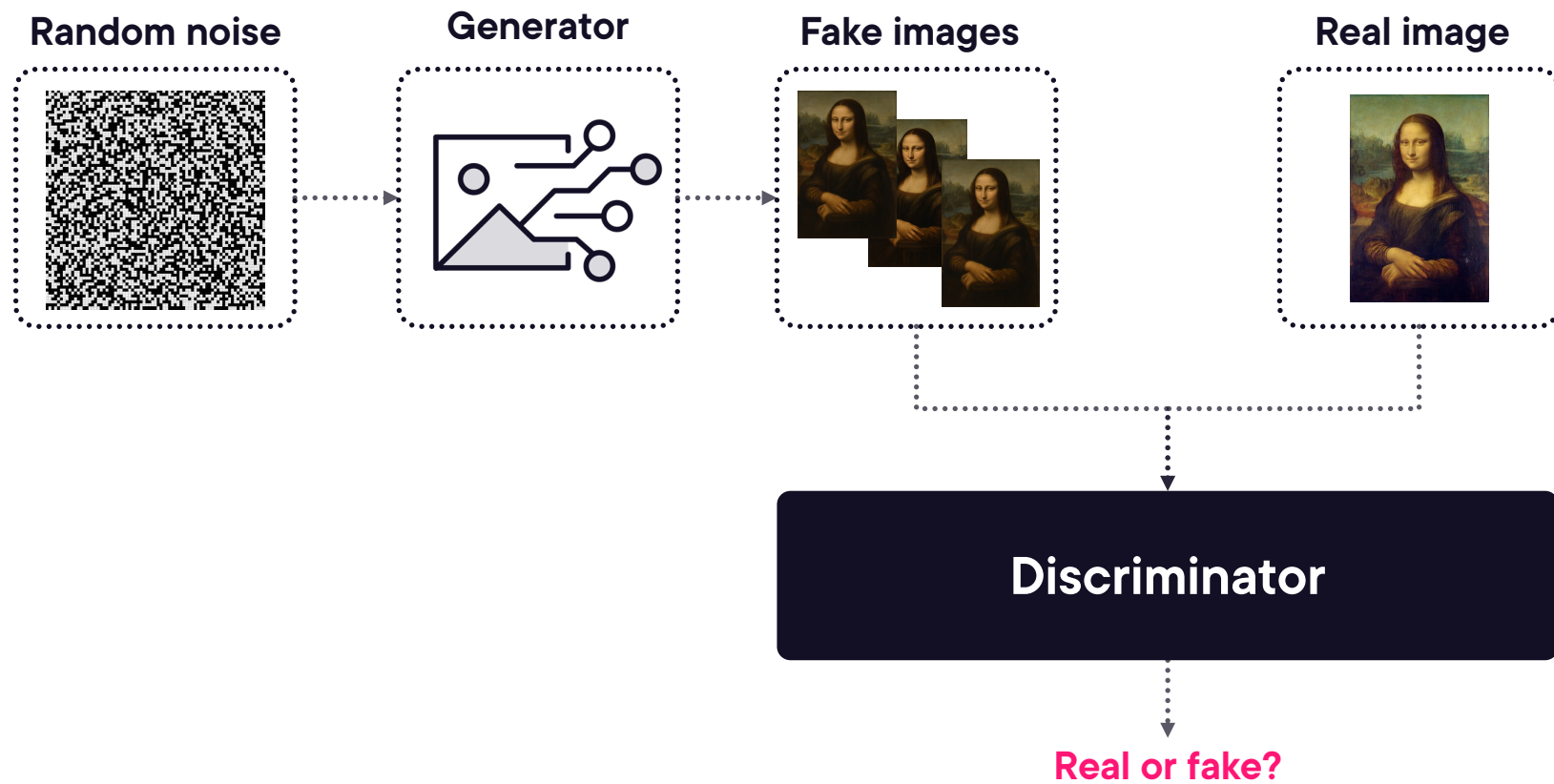


Generator
Creates new data



Discriminator
Tries to tell if data is real or fake

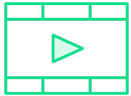




Applications of GANs



Image style transfer



Video generation



Data augmentation



Anomaly detection





Variational Autoencoders (VAEs) Explained



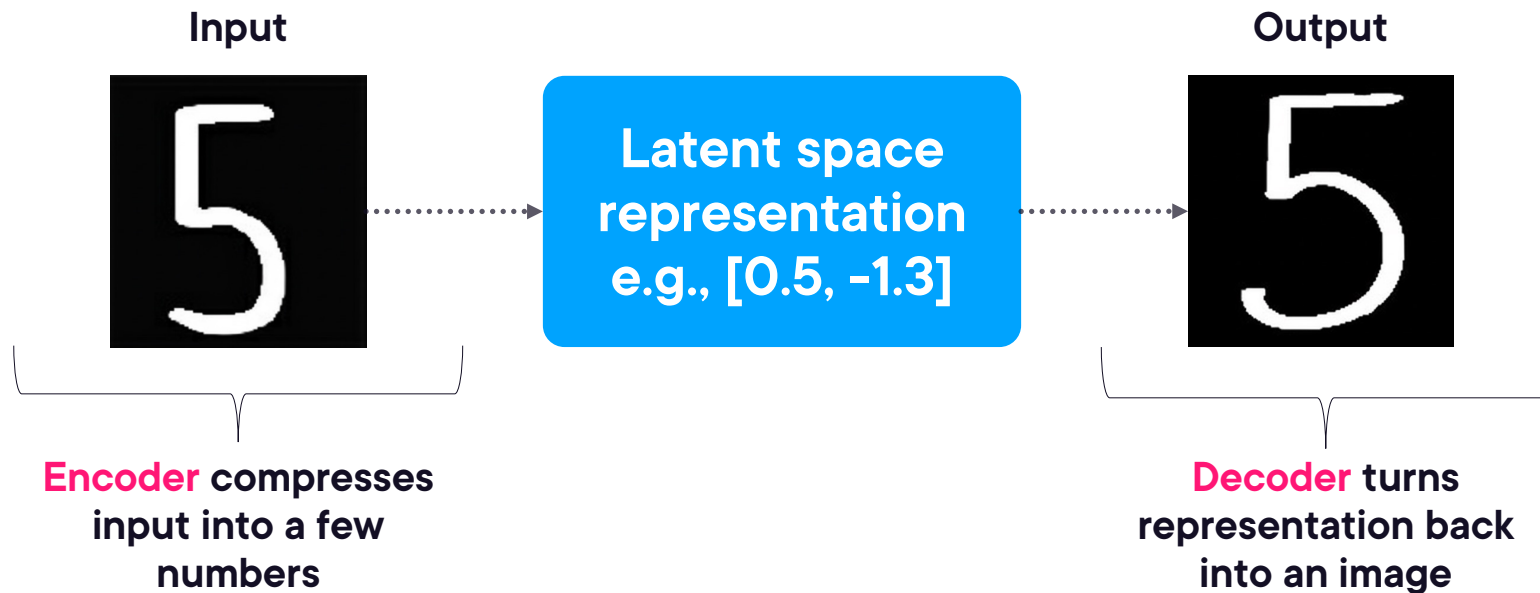
TO:

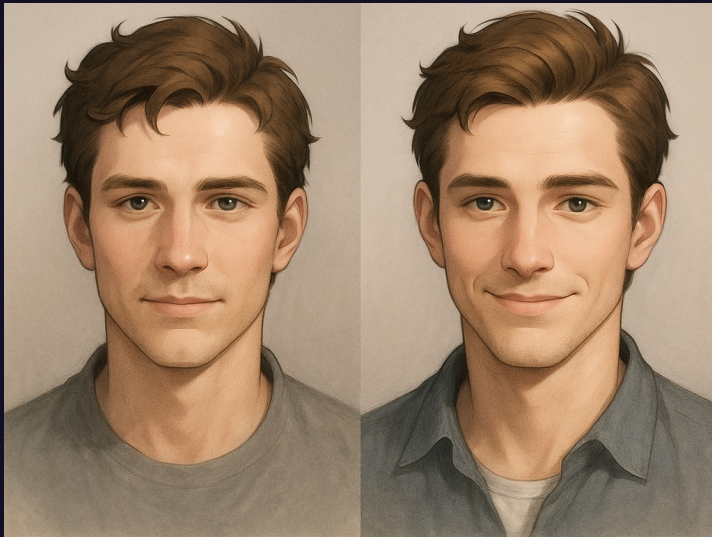
Jamie Walker
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62704



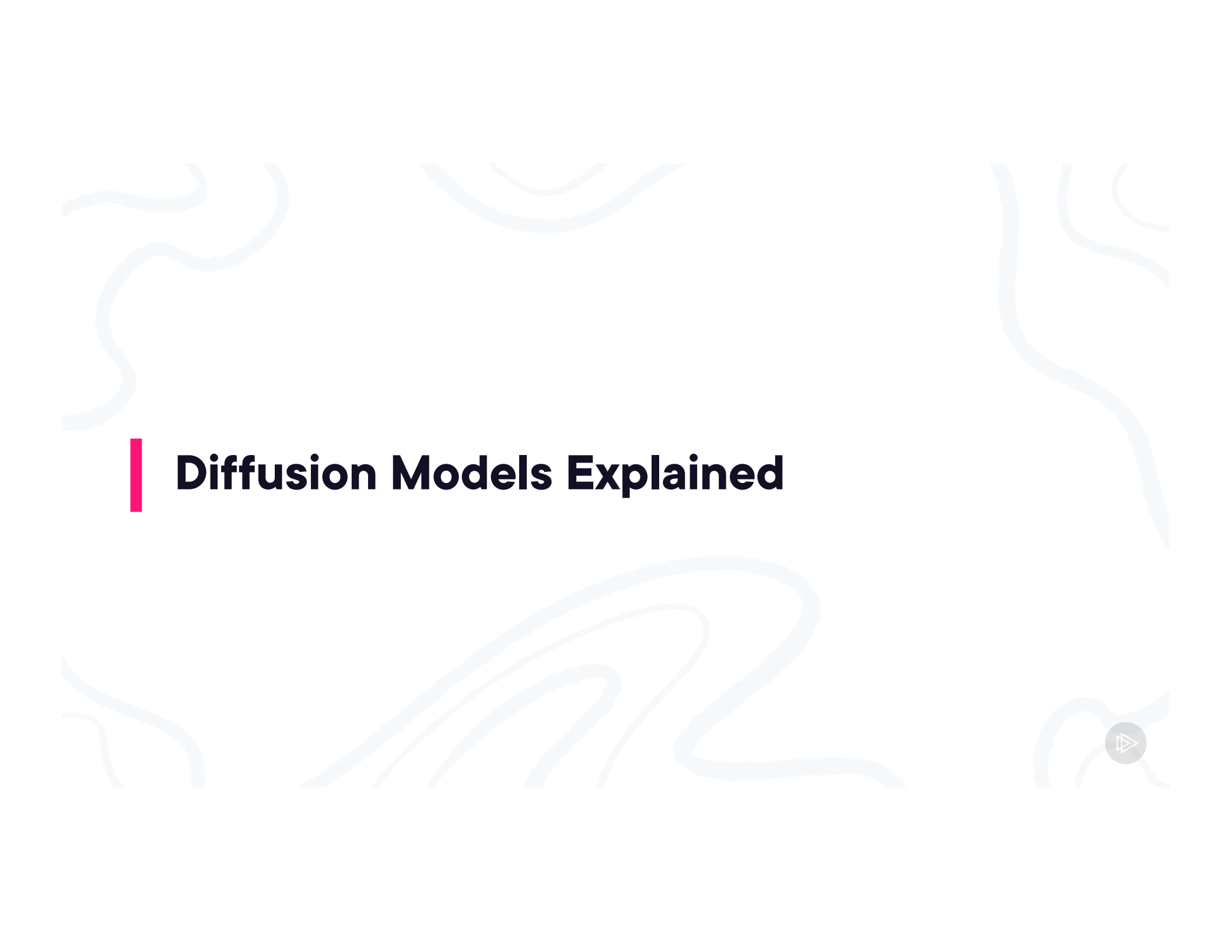
Variational Autoencoders (VAEs)





Generated using ChatGPT





Diffusion Models Explained



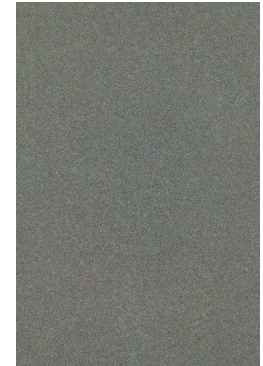


Generated using ChatGPT



Forward diffusion

Gradually adding noise to an image until it's pure static



Generated using ChatGPT

Reverse diffusion

Step-by-step removing noise to recreate a realistic image



Prompt: A backpack sitting on the grass with a bench nearby



Generated using ChatGPT



Diffusion Models vs. GANs and VAEs

MODEL	STRENGTHS	DRAWBACKS
Diffusion Models	• Extremely high-quality, detailed outputs • More stable training • Good at diverse, realistic variations	• Slower to generate images (many steps) • Computationally intensive
Generative Adversarial Networks (GANs)	• Fast image generation • Sharp and visually appealing results	• Harder to train (unstable) • Prone to mode collapse (limited diversity)
Variational Autoencoders (VAEs)	• Simple and stable to train • Good for smooth interpolation and variation	• Blurry outputs • Lower visual quality





Combining Inputs: How Multimodal Models Work





Multimodal Models in Action

Exploring Google AI Studio and OpenAI's Sora

